

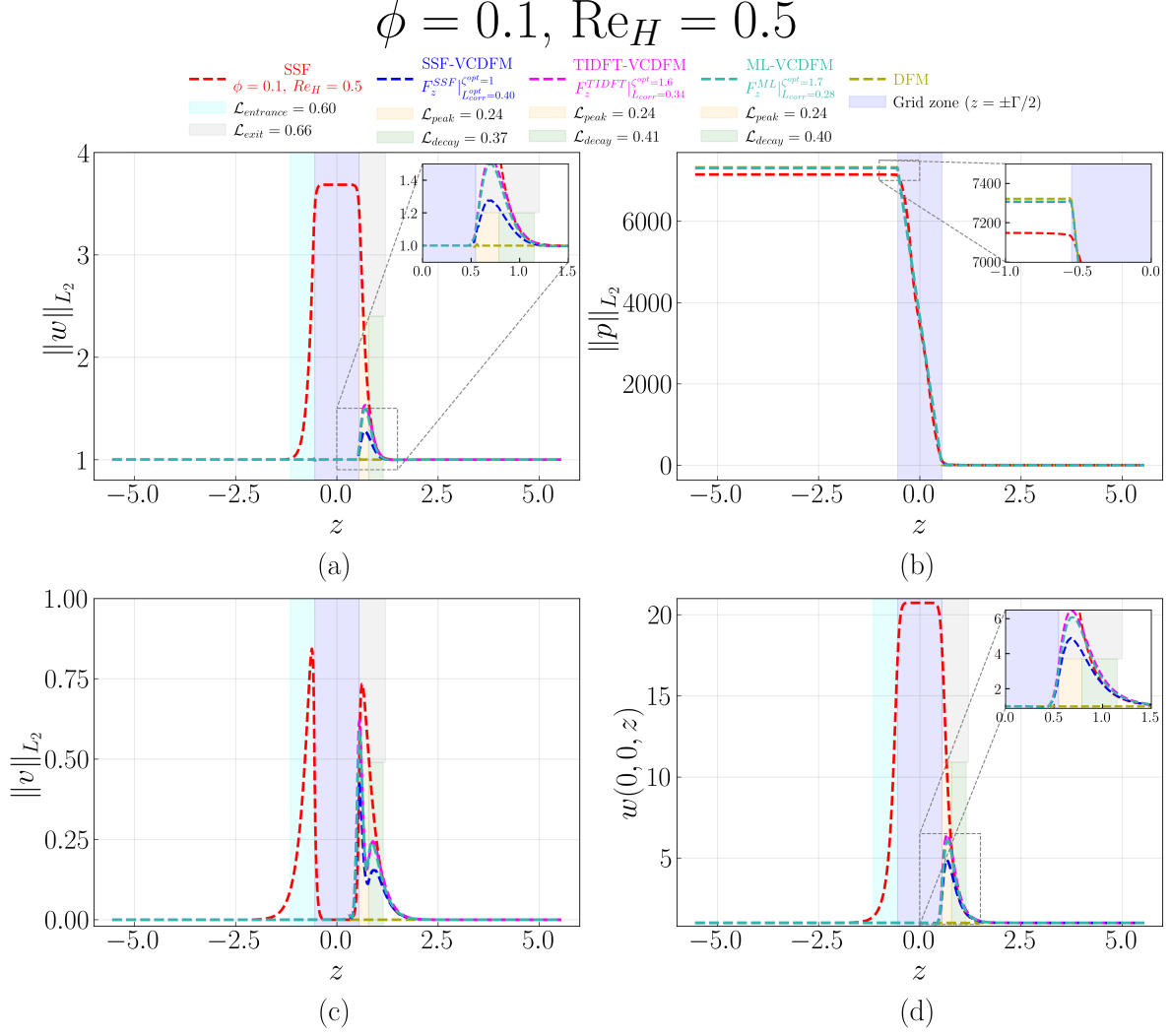


Figure D.27: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.5$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

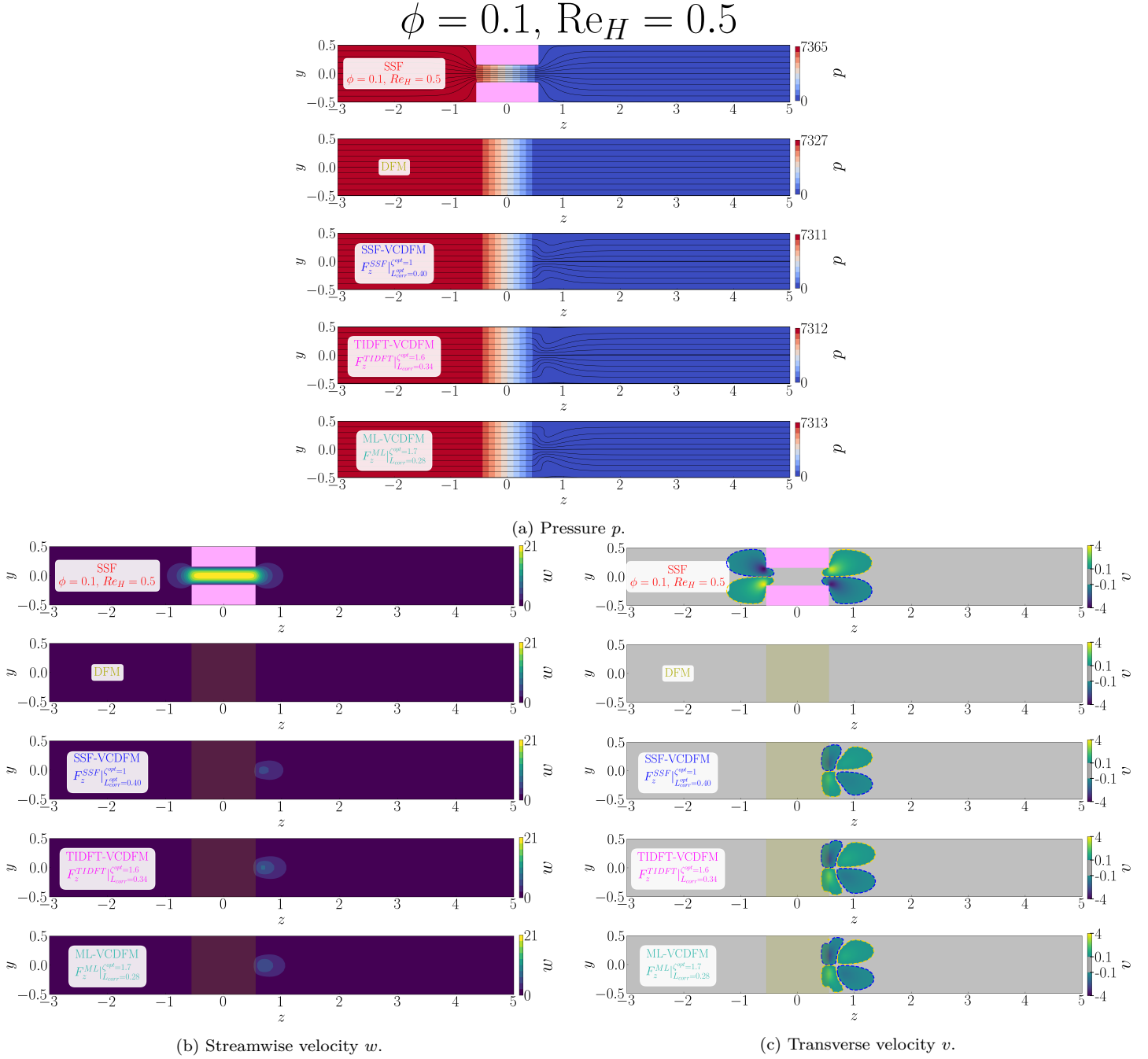


Figure D.28: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $\text{Re}_H = 0.5$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

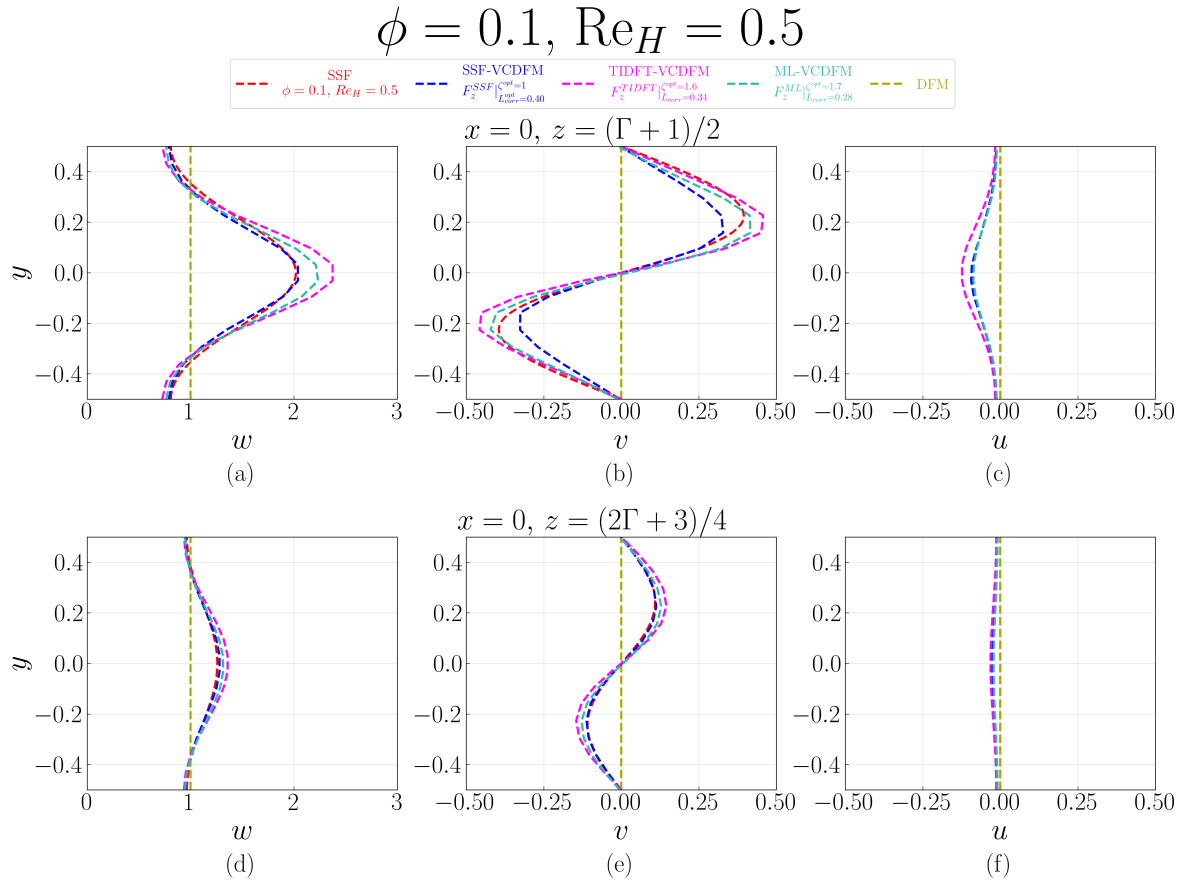


Figure D.29: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.5$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 1)/2$ in (a–c) and $z = (2\Gamma + 3)/4$ in (d–f).

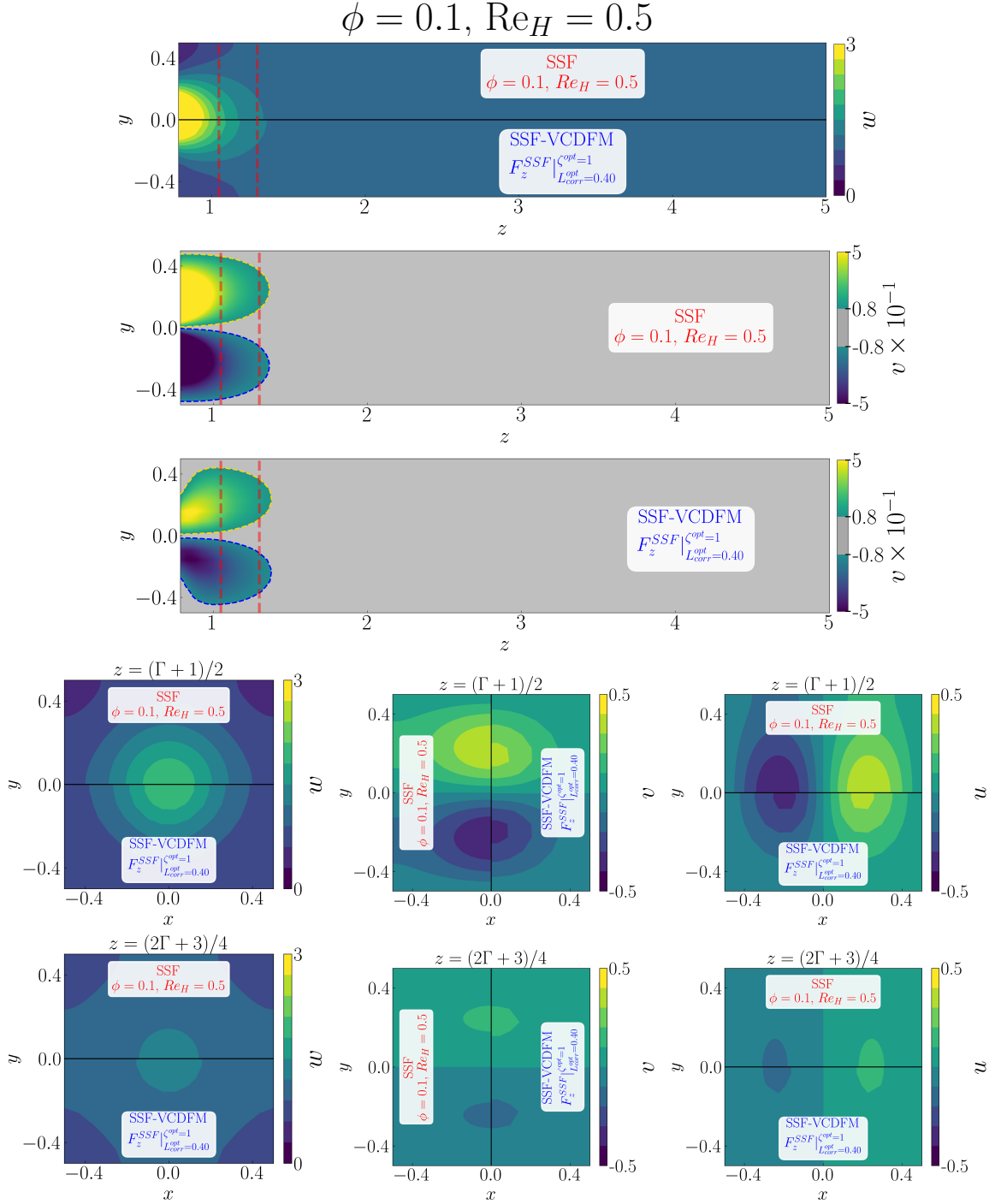


Figure D.30: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.5$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

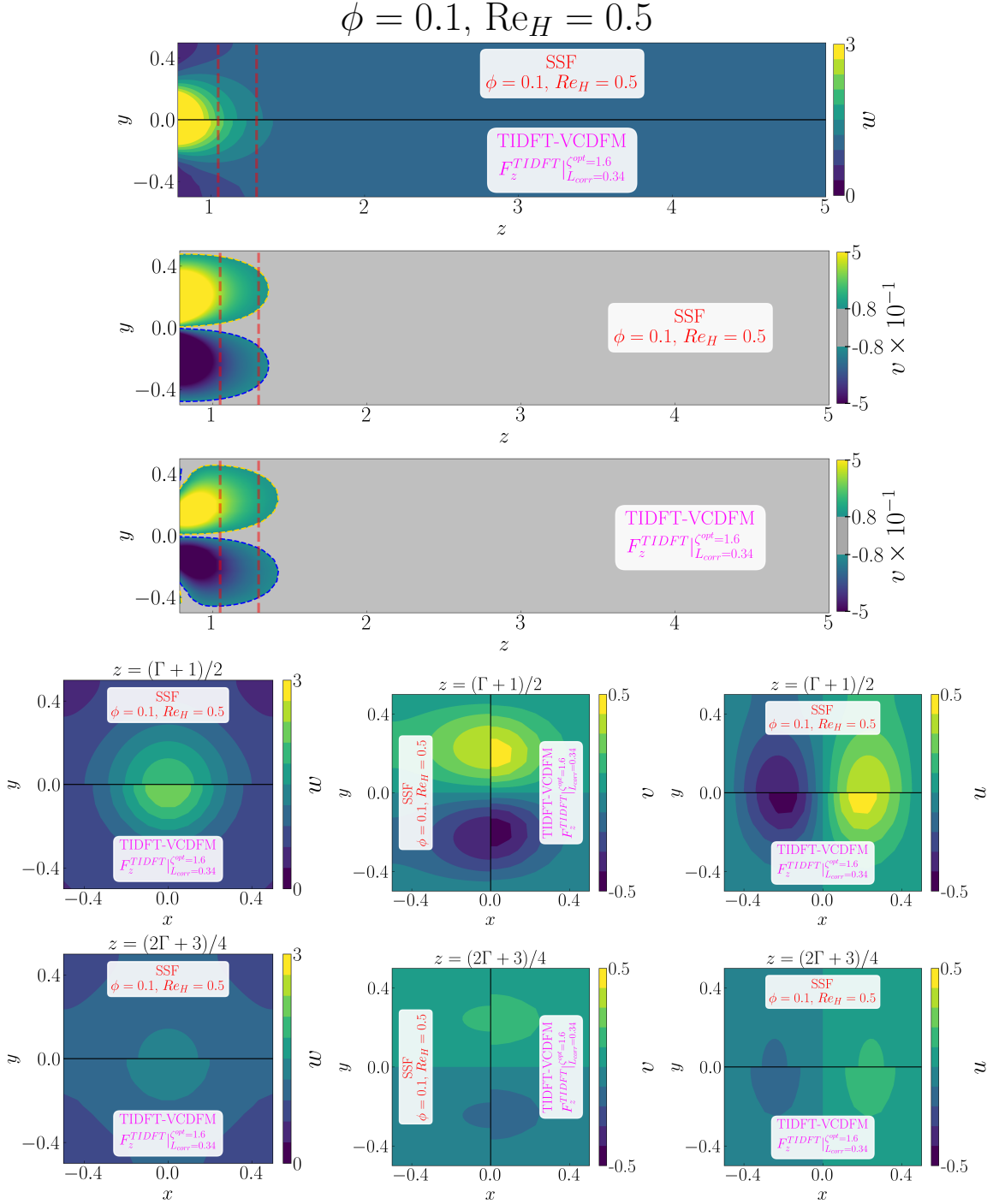


Figure D.31: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.5$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

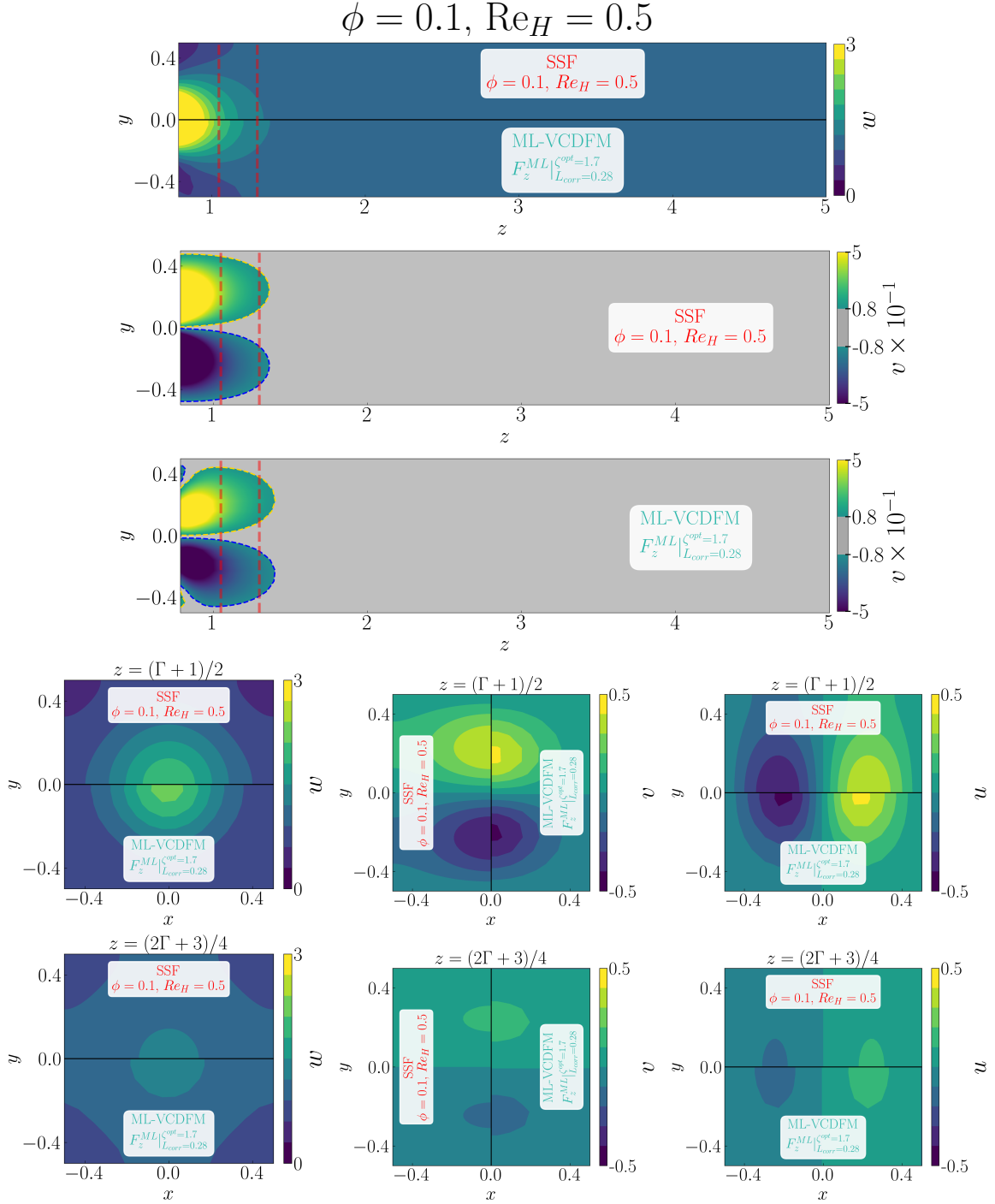


Figure D.32: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.5$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

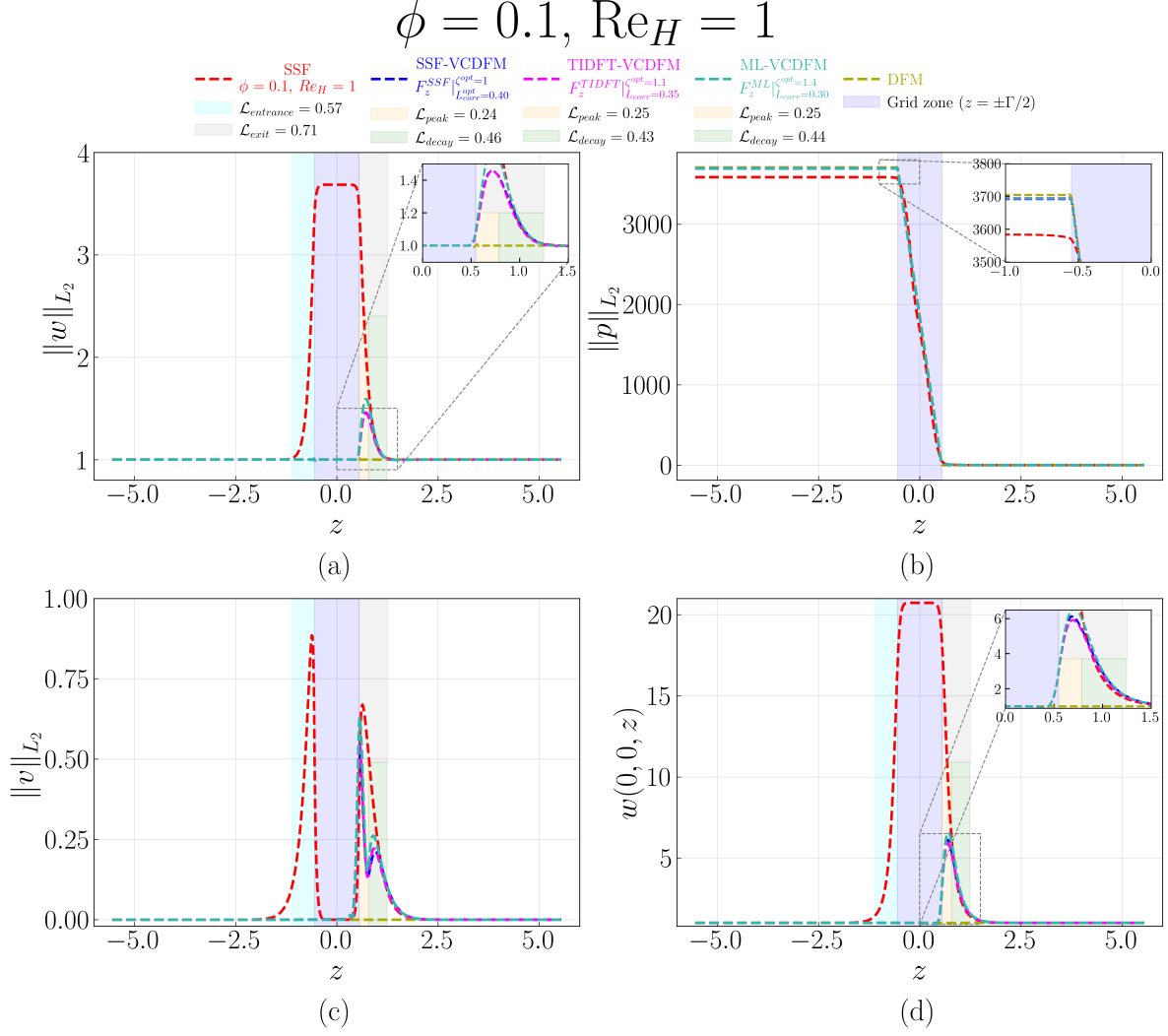


Figure D.33: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 1$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

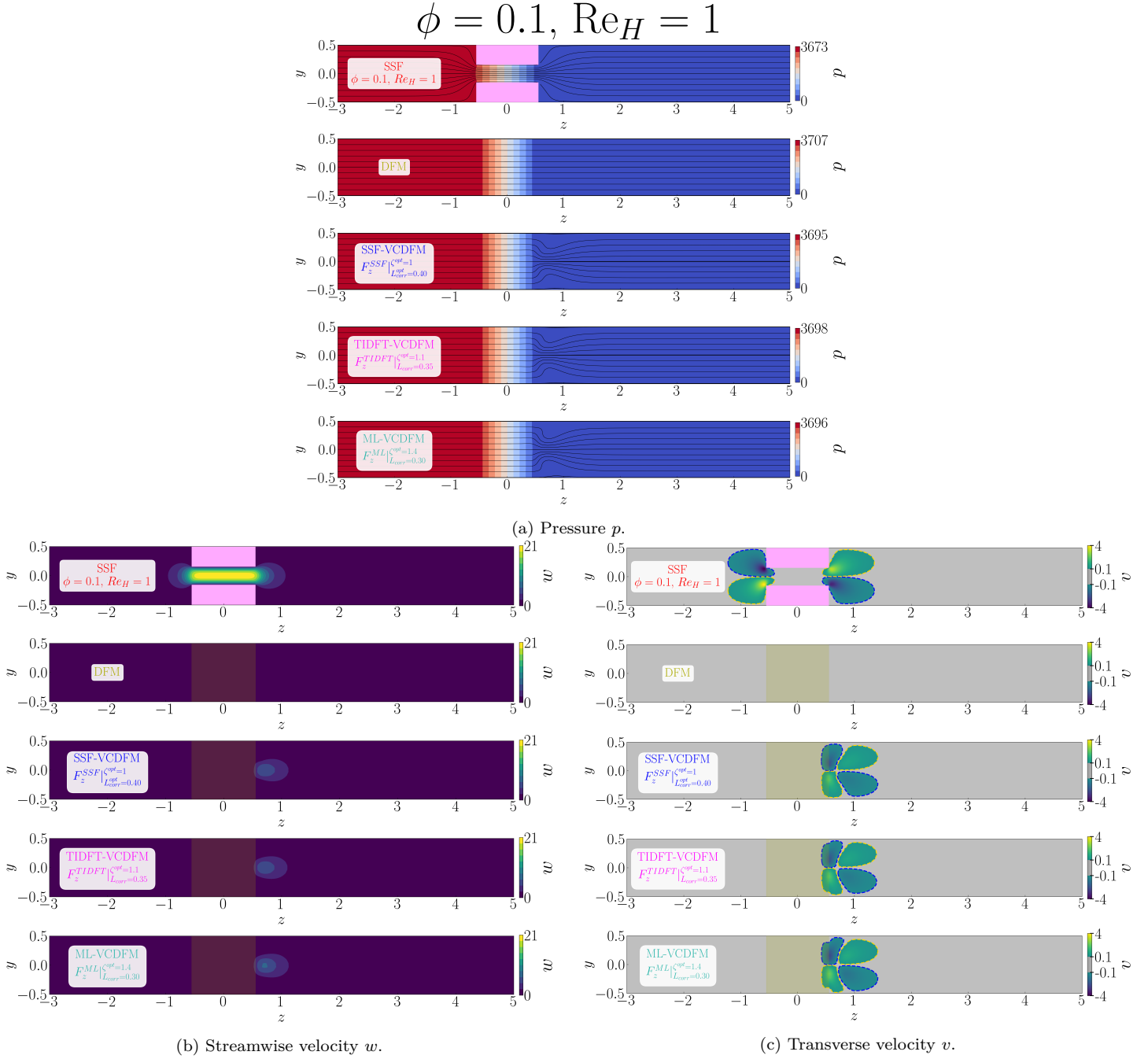


Figure D.34: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $\text{Re}_H = 1$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

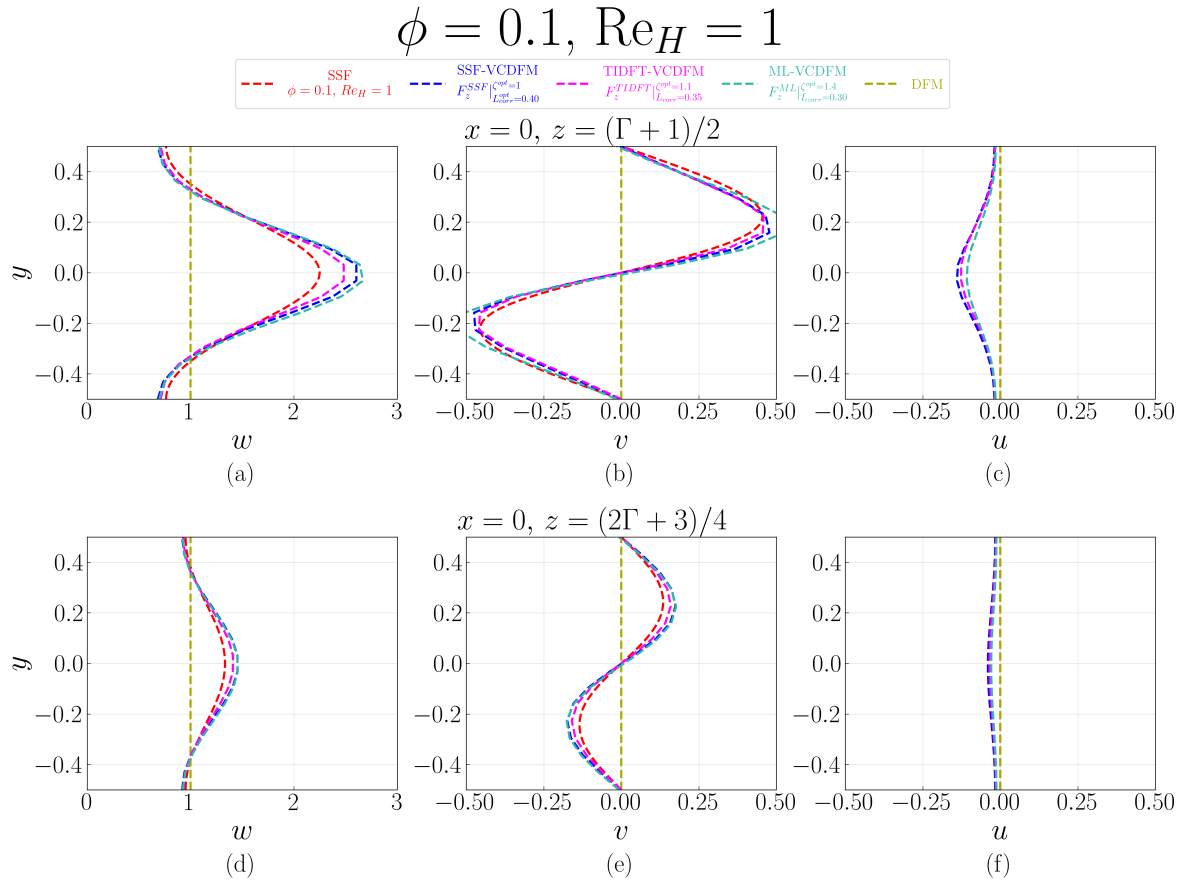


Figure D.35: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 1$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 1)/2$ in (a–c) and $z = (2\Gamma + 3)/4$ in (d–f).

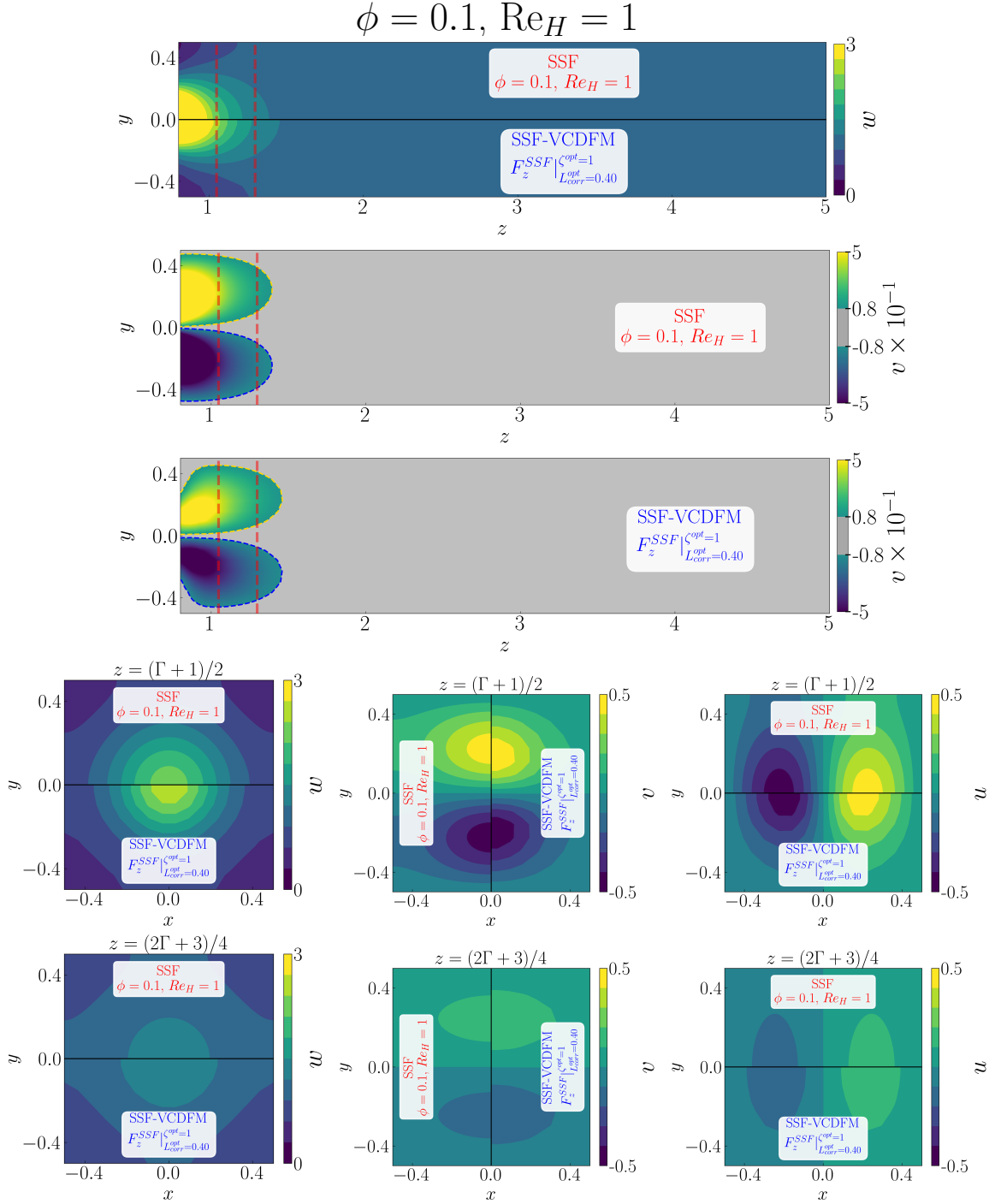


Figure D.36: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 1$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

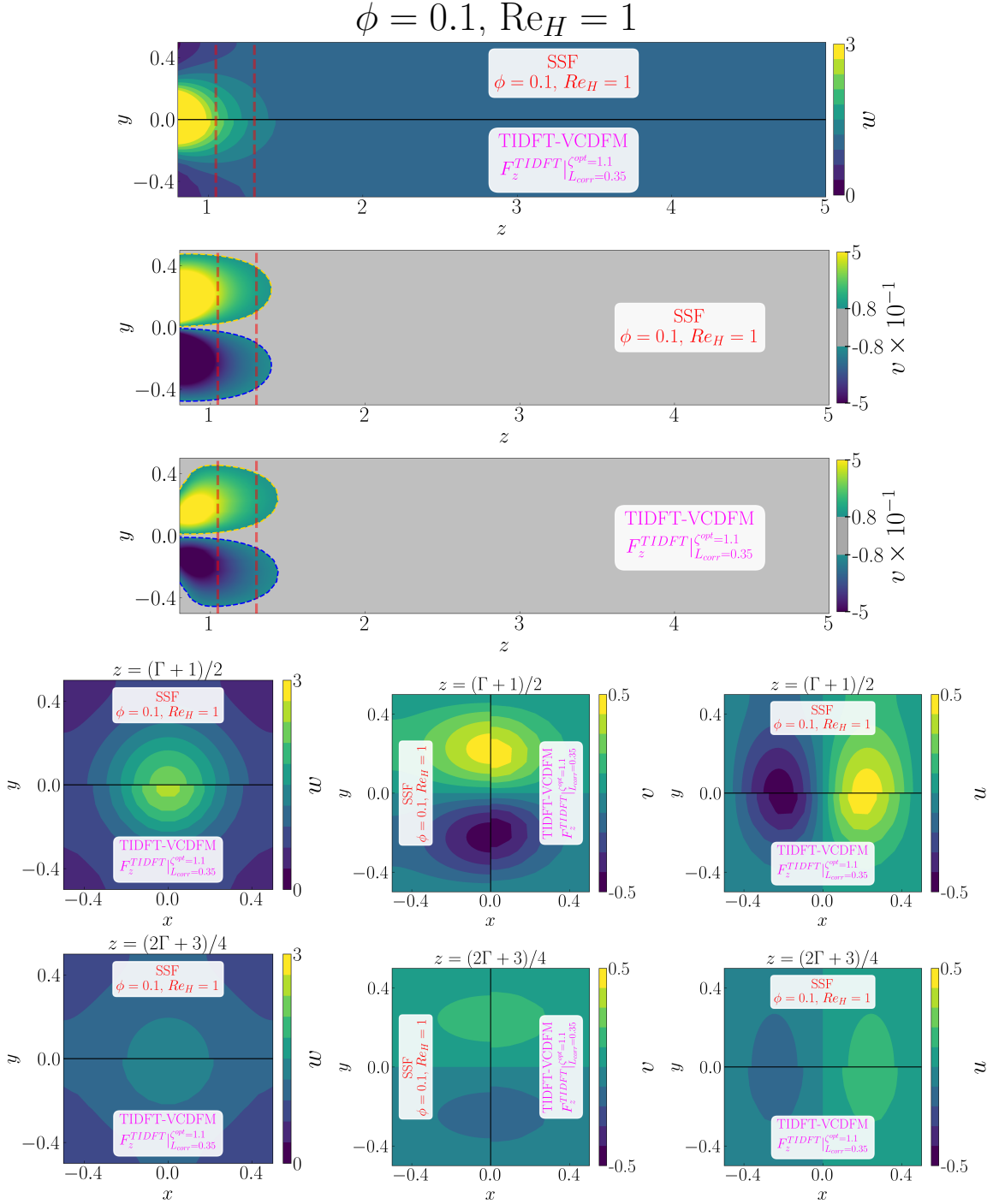


Figure D.37: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 1$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

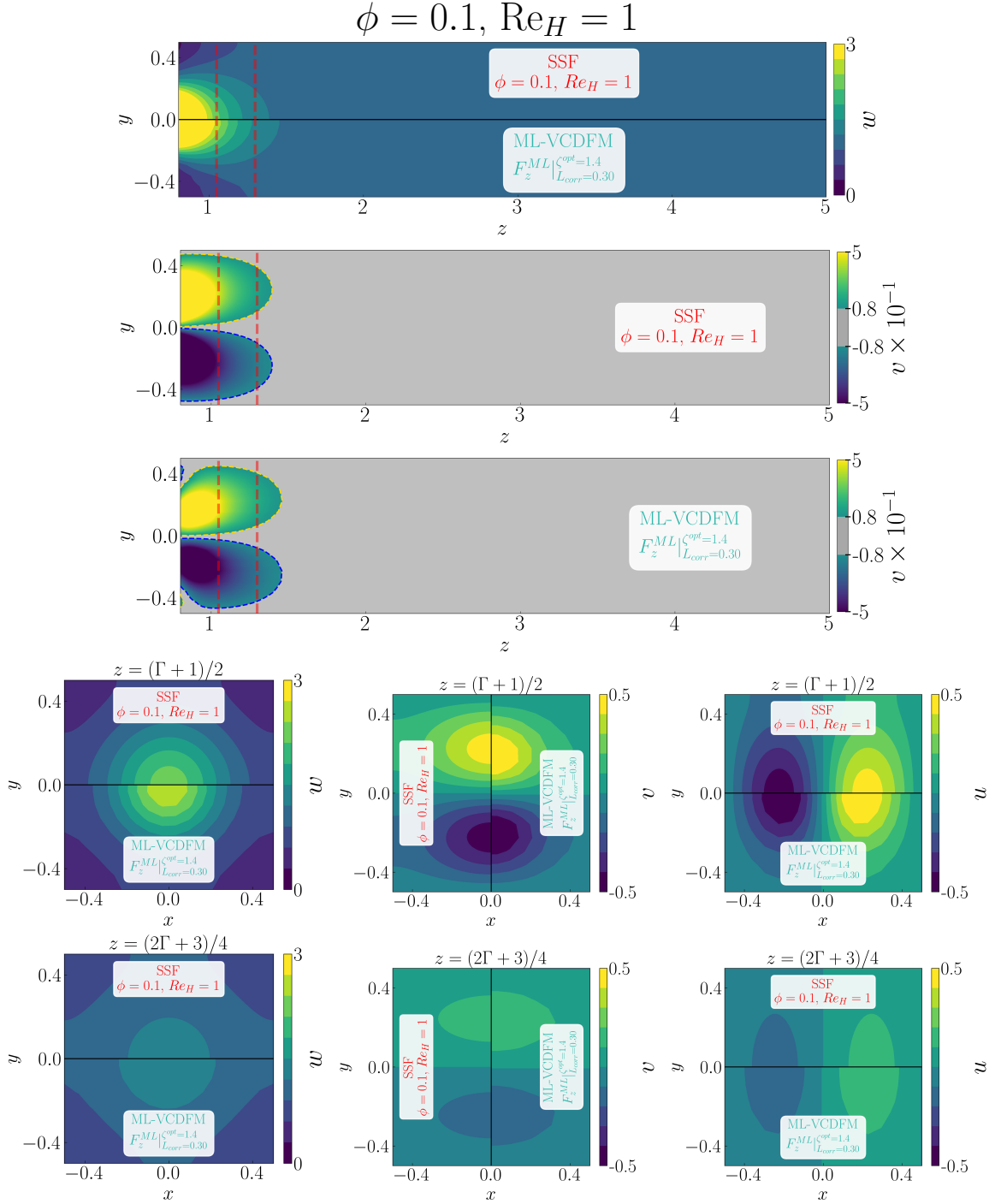


Figure D.38: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 1$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

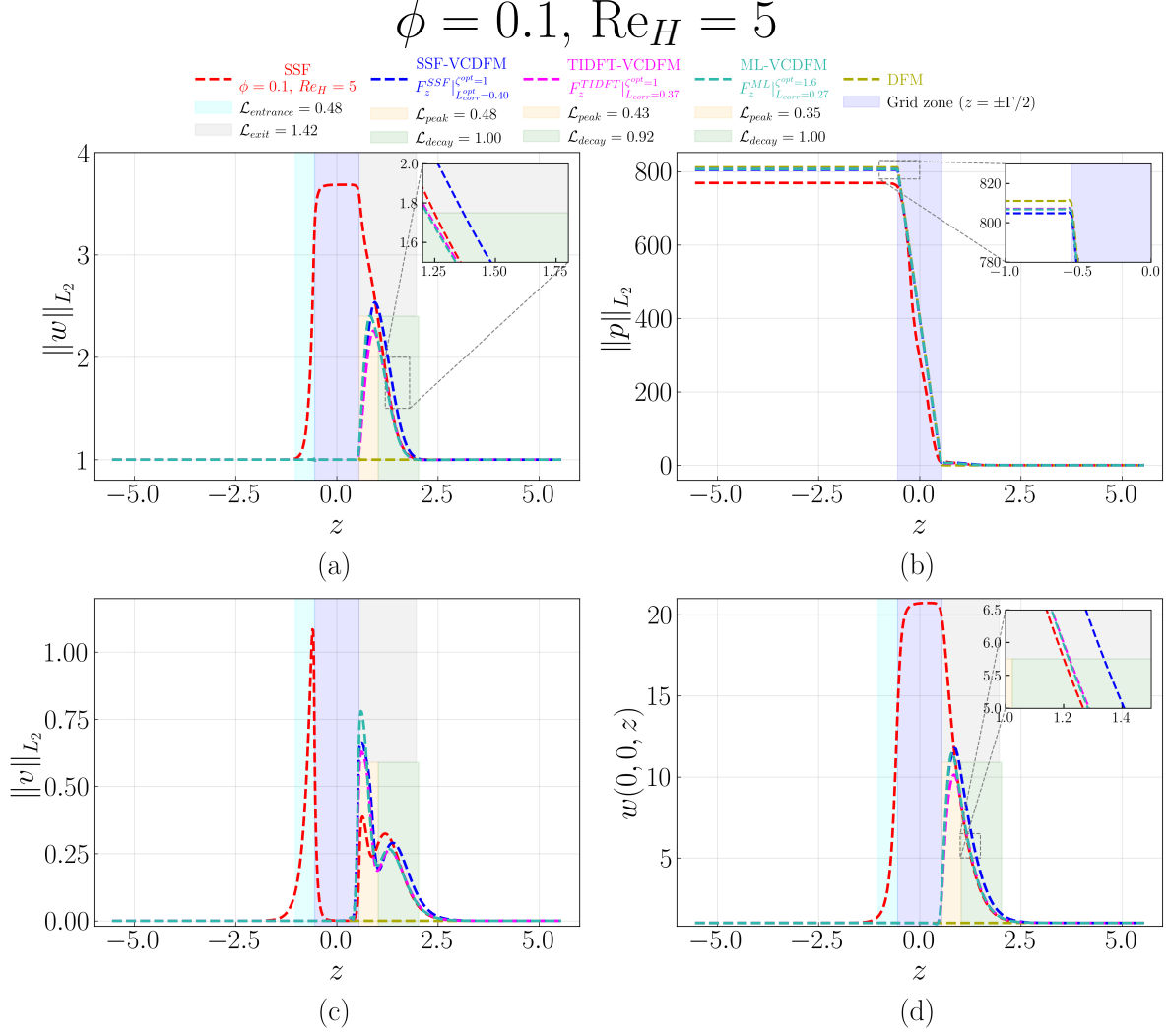


Figure D.39: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 5$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

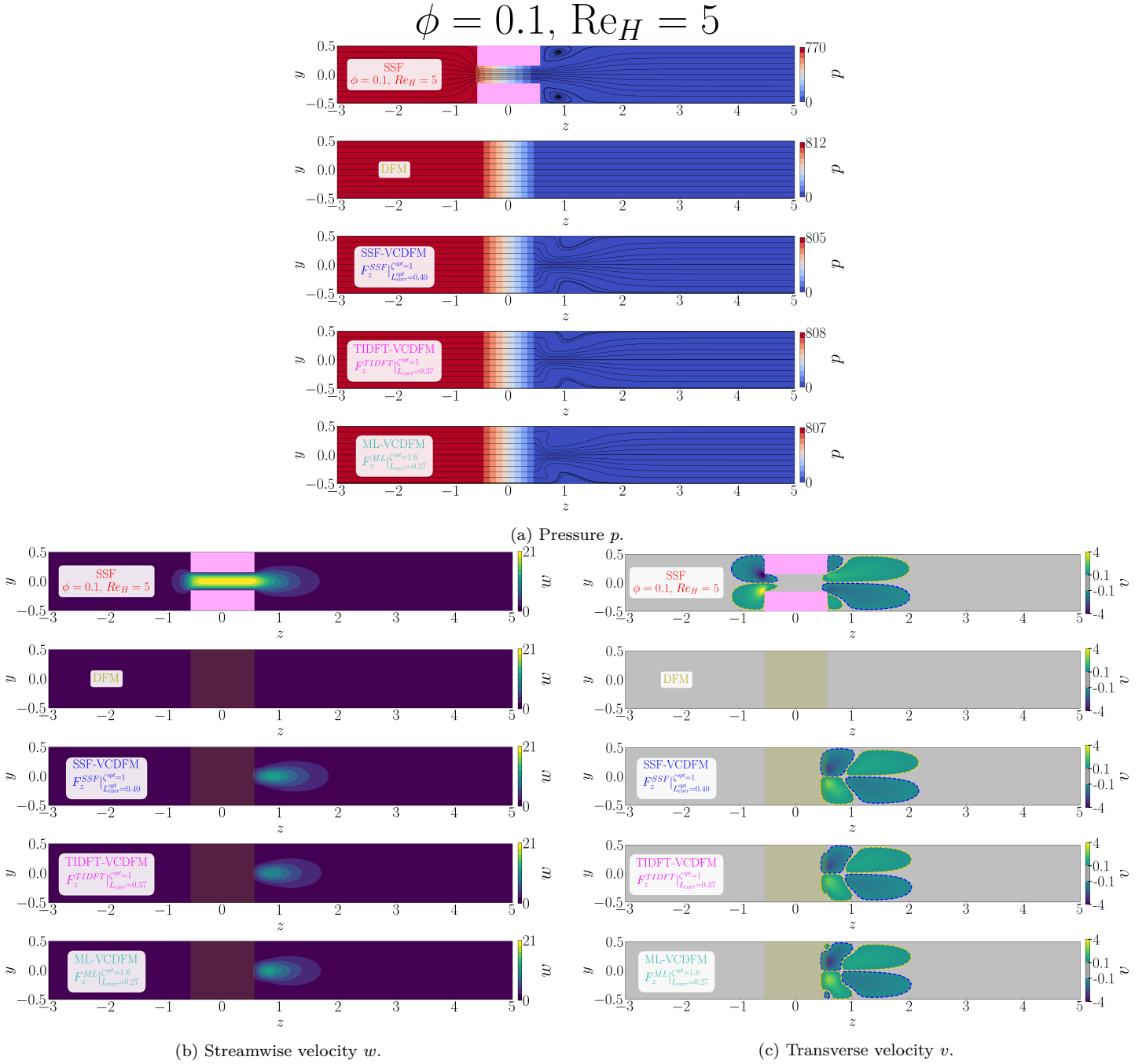


Figure D.40: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $Re_H = 5$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

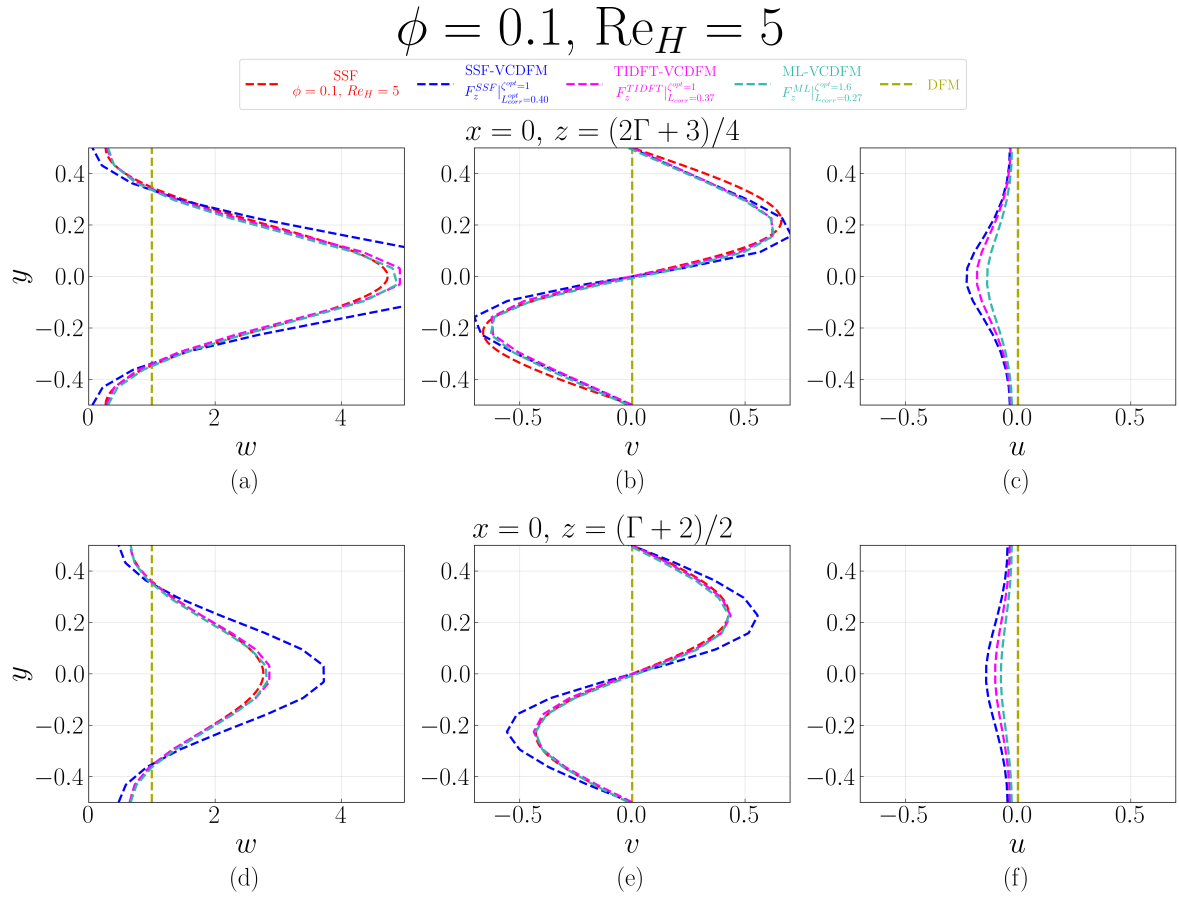


Figure D.41: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 5$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (2\Gamma + 3)/4$ in (a-c) and $z = (\Gamma + 2)/2$ in (d-f).

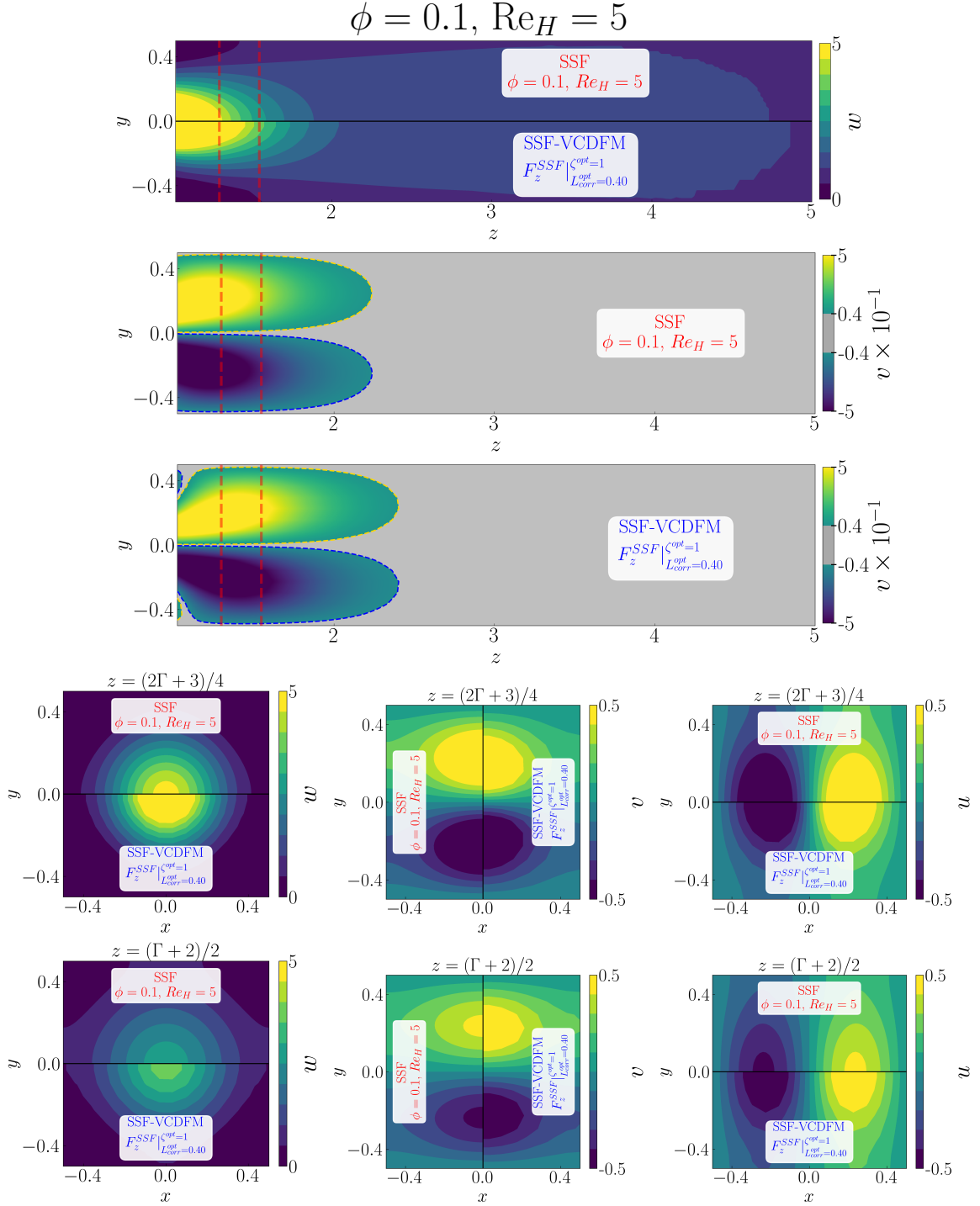


Figure D.42: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 5$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 3)/4$ in the first row and $z = (\Gamma + 2)/2$ in the second row.

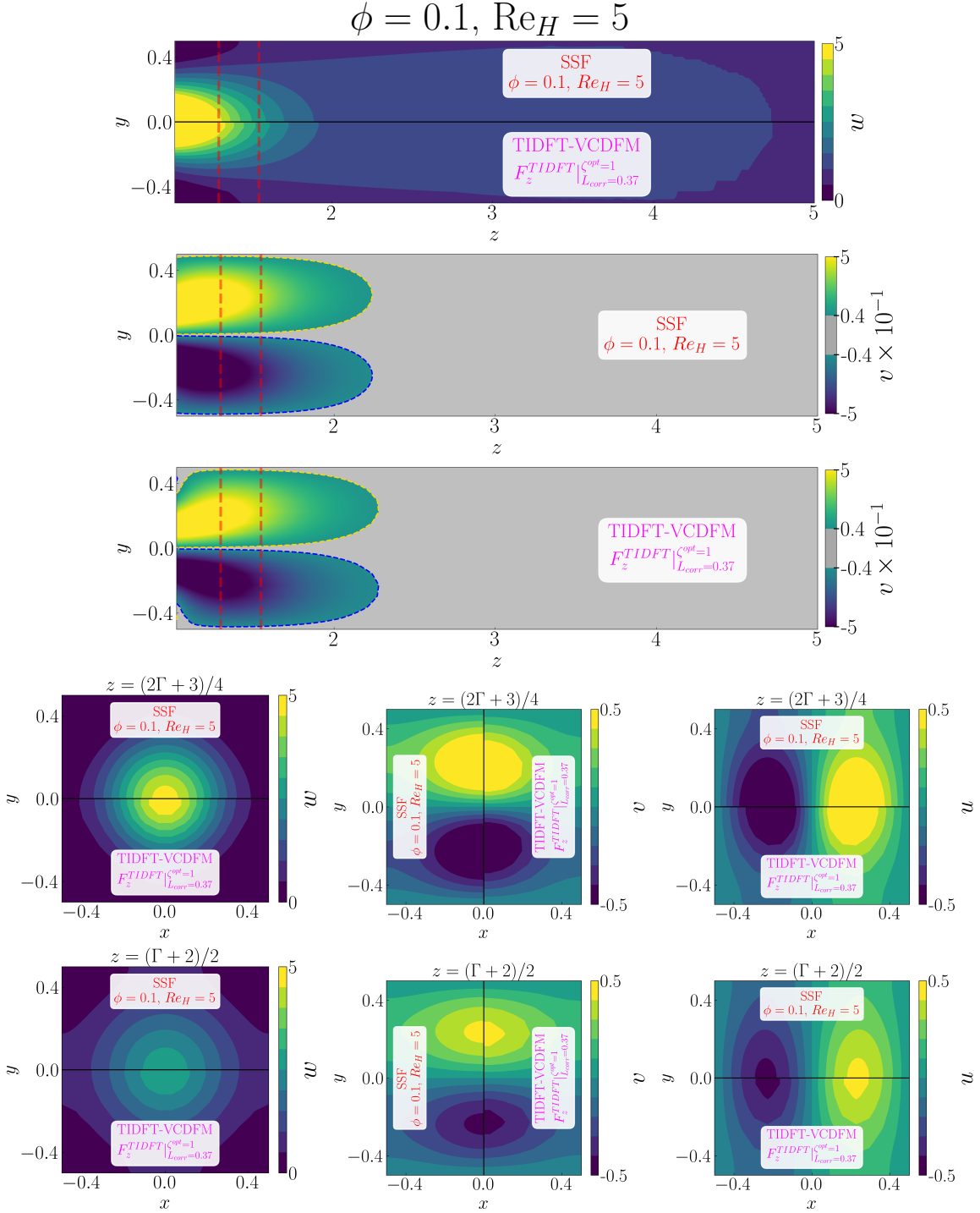


Figure D.43: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 5$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 3)/4$ in the first row and $z = (\Gamma + 2)/2$ in the second row.